

Hamsika Rajeshwar Rao Garugula

hamsikaraog@gmail.com | (716)-518-0089 | [linkedin.com/in/hamsikarg](https://www.linkedin.com/in/hamsikarg) | github.com/HamsikaRaj

AI Engineer | LLM Applications, Agentic Workflows, and Evaluation

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript

LLMs & NLP: Large Language Models, AI Agents, Agentic Workflows, Workflow Automation, Multi-Agent Orchestration, Tool Use / Function Calling, Model Context Protocol (MCP), RAG, Prompt Engineering, LangChain, LangGraph, Context Synthesis, Transformers, Hugging Face, Anthropic / OpenAI APIs

Evaluation & Responsible AI: Evaluation Frameworks, Evaluation Pipelines, LLM-as-Judge, RAGAS, Guardrails, Responsible AI, Human-in-the-Loop, Hallucination Detection, MLOps / LLMOps, Monitoring, Search / Retrieval, Vector Databases (FAISS, Qdrant, ChromaDB, Weaviate)

Engineering: FastAPI, Flask, REST APIs, PyTorch, TensorFlow, Scikit-learn, Docker, AWS, GCP, Git, CI/CD

EXPERIENCE

AI Engineer | Saayam for All

February 2026 – Present

- Built a multi-model LLM assistant routing user queries across **4 providers** (Grok, ChatGPT, Gemini, Meta AI) via a zero-shot BART classifier that predicts query category and dispatches to the right model. Flask backend, live end-to-end from problem framing to deployment.
- Debugged an **18x latency regression** in production by controlling for output length to disprove the obvious model-size hypothesis, isolating the real cause to an unsupported API client, then set the routing policy that removed the slow provider.
- Built an automated quality-check layer scoring every generated answer on accuracy and clarity, turning output review from a manual spot-check into a gated, measurable step across all 4 providers.

AI & ML Engineering Intern | L&T Technology Services

June 2023 – August 2023

- Engineered a production OCR pipeline in Python with automated preprocessing across **18+ regional plate formats**, achieving 85% format compatibility and reducing error rates by 25% at 300+ DPI.
- Optimized the image preprocessing stack (normalization, skew correction, noise removal with OpenCV), improving OCR accuracy by **30%** across diverse regional formats.

Software Engineering Intern | Kleza Solutions Pvt Ltd

July 2022 – March 2023

- Built and deployed backend REST APIs and microservices for a production healthcare Care Assessment App (Spring Boot, PostgreSQL relational + MongoDB NoSQL), cutting API response times by **50%** through query optimization.
- Deployed an XGBoost model into the live production pipeline for clinical risk scoring, automating high-risk patient flagging and follow-up routing that had been manual review work.
- Partnered with a clinical (non-engineering) team across the full development lifecycle and wrote the technical documentation they relied on.

PROJECTS

Multi-Agent Coding Assistant | *Python, FastAPI, LangGraph, Anthropic API, React, TypeScript, Tailwind, Vite, SSE*

- Architected an AI application as a 5-agent LangGraph pipeline (Planner to Architect to Coder to 2x Reviewer) that generates multi-file codebases from one prompt, shipping a **25-file React/TypeScript app in a single run**, with an automated review loop (up to 3 retries) that caught and repaired broken output before it shipped.
- Cut LLM inference cost by routing per task across model tiers: Opus for generation (~82% of per-run cost), Sonnet for planning/review (a full dual-reviewer cycle ran **\$0.0007**), with a Haiku intent classifier fronting both code-gen and chat. Streamed per-agent telemetry over Server-Sent Events (SSE), surfacing the bottleneck (~8.7s coder vs ~1.5s reviewer).

RAG Evaluation Harness | *Python, FastAPI, FAISS, Claude API, Sentence Transformers, Pydantic, Pytest*

- Built an automated **LLM evaluation pipeline (LLM-as-judge)** that defines and scores quality metrics (faithfulness, answer relevancy, hallucination) to gate changes before release, cutting evaluation effort ~90%. On a 75-question benchmark: faithfulness 0.996, answer relevancy 0.82, hallucination 0.017 (98 claims verified, 0.98 grounding).
- Built a pluggable multi-backend retrieval layer (FAISS, ChromaDB, Qdrant, Weaviate) behind a FastAPI gateway, validated by a **147-test pytest suite**, and benchmarked 75 queries at Recall@5 0.87 / Recall@10 0.92, MRR 0.75.

RAG Chatbot with Role-Based Access Control (RBAC) | *Python, FastAPI, Streamlit, Qdrant, Claude Sonnet, Sentence Transformers, JWT*

- Designed a RAG system with dual-layer RBAC across 6 role tiers and 5 department knowledge bases, reaching **100% access-control accuracy and 0 hallucinations** across 16 adversarial test cases.
- Built a JSON Web Token (JWT) auth pipeline deriving employee roles dynamically from a 100-person HR dataset across 13 departments, and an ingestion pipeline turning 9 sources into 225 metadata-tagged vectors at sub-second retrieval.

Sentiment Analysis on Customer Reviews | *Python, NLTK, Word2Vec, LSTM, TensorFlow*

- Built an NLP pipeline preprocessing text with NLTK, TF-IDF, and Word2Vec, comparing approaches from a Naive Bayes baseline (58%) forward.
- Implemented an LSTM RNN in TensorFlow, improving accuracy to **80%**, a 22% lift over the baseline.

EDUCATION

M.S. Artificial Intelligence

University at Buffalo, SUNY

August 2024 – May 2026

Buffalo, New York

B.E. Electronics & Communication Engineering

Chaitanya Bharathi Institute of Technology

December 2020 – May 2024

Hyderabad, India

PUBLICATIONS

Survey on Antennas for Different Cancers and Tumor Detection, ISDA 2023, Springer. doi.org/10.1007/978-3-031-64776-5_31